Kevin Abate

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Education

Politecnico di Milano — M.Sc. Computer Science and Engineering – Data Science & Software Engineering — Sep 2024 – Apr 2027
Artificial neural network & deep learning, Recommender system, Machine learning, Design and implementation of mobile application, Software engineering II.

Politecnico di Milano — B.Sc. Computer Science Engineering — Sep 2021 – Jul 2024
Data Structures & Algorithms, Probability & Statistics, Software Engineering (Java), Linear Algebra

Experience

Graduate Student Researcher – NECSTLab – Politecnico di Milano — Jun 2026 – Present — Milan, IT
- Extending a **Multi-Agent System** for autonomous **Root Cause Analysis** (RCA) in Kubernetes
- Designing a **Deep Learning model** for multivariate anomaly detection on Prometheus metrics, integrated into the agent pipeline via a custom **Model Context Protocol** (MCP) tool that distills raw time-series into semantic signals, reducing context window saturation.

Teaching Assistant – Politecnico di Milano — Sep 2025 – Dec 2026 — Milan, IT
- **Informatica A** (Sep–Dec 2026): mentored 100+ undergraduates in procedural programming (C).
- **Algoritmi e Principi dell'Informatica** (Jun–Sep 2026): project tutor guiding students through algorithm design and theoretical computer science concepts.
- **Informatica B** (Sep–Dec 2025): mentored 100+ undergraduates in procedural programming (C) and numerical computing (MATLAB); led weekly lab sessions debugging code and clarifying algorithmic concepts.

Selected Projects

DivAid – Expense-Splitting Mobile App — Mar 2026 – Jun 2026
- Built a full-stack Flutter app (Android/iOS/Web) for shared expense management: group creation, multi-currency balances, greedy debt-minimisation algorithm, and Stripe-powered in-app wallet settlements.
- Architected a feature-first, layered codebase (BLoC/Cubit state management, Cloud Functions in TypeScript for privileged ops); integrated Groq LLaMA 4 Vision for receipt OCR and ECB live exchange rates.
- *Stack*: Flutter/Dart, Firebase (Firestore, Auth, FCM, Cloud Functions), Groq API, Stripe API.

Deep Learning – Medical Image & Time Series Classification — Oct 2025 – Jan 2026
- Led a team of 4 to **top 5%** (F1 = 0.4469) in Medical Imaging classification and **top 20%** (F1 = 0.92) in Time Series classification among ∼200 Kaggle teams.
- Applied Transfer Learning on domain Vision Transformers (UNI, Virchow2), Hybrid CNN-LSTM with Attention, CatBoost ensembling, and Optuna hyperparameter search to overcome severe data scarcity.
- *Stack*: PyTorch, TensorFlow/Keras, Optuna, CatBoost, XGBoost, scikit-learn.

Recommender System – Large-Scale Hybrid Stacking — Oct 2025 – Jan 2026
- Co-led a 2-person team to **7th place / 70+ teams** on an implicit feedback dataset (27k users, 7k items).
- Built a 2-stage ensemble: collaborative filtering for candidate generation re-ranked with **XGBoost**; optimized hot paths via **Cython** for low-latency inference.
- *Stack*: Python, XGBoost, Cython, scikit-learn, pandas.

Skills and Languages

Languages	Python, Java, C/C++, Dart/Flutter, SQL
AI / ML	PyTorch, TensorFlow/Keras, scikit-learn, Optuna, pandas, NumPy
Tools	Git/Github, Docker, Kubernetes
Databases	MySQL, MongoDB, Neo4J, Redis
Spoken Languages	Italian (native); English C1

Volunteering

Google Developer Group (GDG) – PoliMi — Sep 2025 – Present — Milan, IT
- Co-organised **GDG AI Hackathon**, a competitive AI event drawing participants from 10+ countries; connect engineering students with industry professionals through technical workshops.